Unit 3 - Important Topics

Grain configurations, burning types, burn rate factors, mass generation relation aur solved numericals

21ASC305T - Rocket Propulsion
Slides ke basis par simple Hinglish me samjhaya hua notes

Unit 3 - Important Topics (Solid Rocket)

Ye PDF sirf un topics ka hai jo important bataye gaye hain. Har topic ke end me **Exam Answer** (English me, full marks ke liye) diya hai. Last me numericals hain.

Is PDF me kya kya hai

- Solid propellant grain configurations: geometry, advantages, limitations
- Progressive, neutral aur regressive burning
- Burn rate ko influence karne wale factors
- Burning rate, burning surface area aur propellant mass generation rate ka relation
- Slides ke 6 solved numericals (propellant mass, burn rate, chamber pressure, thrust, throat area)

Source ka matlab: hare box me PDF ka naam aur page number (slide number) diya hai. Grain configurations ke topic ka detail repo ke slides me nahi hai (slides me sirf grain design requirements aur ek 5-slot grain ka diagram hai). Isliye us topic ke liye maine standard textbook (G.P. Sutton, Rocket Propulsion Elements, jo syllabus ki reference book 1 hai) ka content use kiya hai, aur wahan source me ye likha hai.

1. Solid propellant grain configurations

Source: Slides me sirf Lecture 1 | Page 13-14 (5-slot grain ka diagram) aur Page 24-28 (grain design requirements). Configurations ka detail: textbook (Sutton, Rocket Propulsion Elements)

Grain propellant ka wo shaped block hai jo motor ke andar jalta hai. Grain ki shape (geometry) hi decide karti hai ki burning surface area A_b time ke saath kaise badlega, aur isliye thrust-time curve kaisa aayega.

Pehle kuch terms:

- **Web thickness (w):** wo propellant thickness jo burning surface ko jalke tay karni hai. Burn time = w / r.
- **Volumetric loading:** chamber ke volume me se kitne hissa me propellant bhara hai. Zyada ho to motor compact hota hai.
- **Slivers:** end me bacha hua unburnt propellant jo bekaar jaata hai (Lecture 1, Page 14).
- **Inhibitor:** propellant ki wo surface jo coating karke jalne se roki gayi hai.

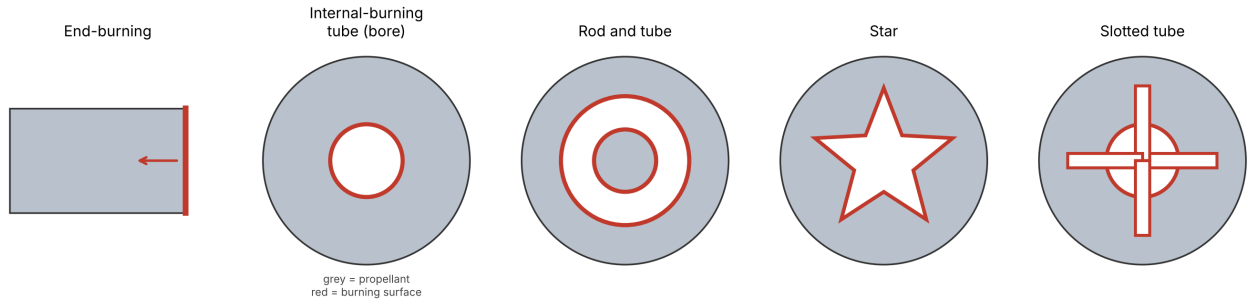


Figure: grey part propellant hai, white part khali cavity hai, aur laal line wo surface hai jo jal rahi hai. End-burning me side view hai (laal line wo end face hai jo jalti hai).

Type	Geometry (kaisi shape)	Burning type	Advantages	Limitations
End-burning (cigarette)	Solid cylinder. Sirf ek end face jalti hai, axial direction me. A_b = pi $D^2/4$	Neutral (A_b constant)	Bahut simple. Constant thrust. Long burn time. Volumetric loading high. Case ke upar unburnt propellant hone se case thanda rehta hai. Slivers kam.	Thrust low (A_b chhota). C.G. shift hota hai. Case ka end hissa garam gas ke contact me aata hai. Grain ko case se bond karna zaroori.
Internal-burning tube (cylindrical bore)	Cylinder ke beech me gol hole (bore). Bore ki surface andar se bahar ki taraf jalti hai. A_b = pi d L	Progressive (bore bada hota hai, A_b badhta hai)	Simple aur common. Thrust high mil sakta hai. Case propellant se protect rehta hai (grain case aur flame ke beech me).	Thrust aur pressure time ke saath badhte hain (progressive). Slivers bachte hain. Bore me gas velocity zyada ho to erosive burning.
Rod and tube	Bahar ek tube aur beech me ek rod. Tube ki andar wali surface aur rod ki bahar wali surface dono jalti hain.	Lagbhag neutral (ek surface badhti hai, doosri ghatti hai)	Chhote size me bada burning area, isliye high thrust. Neutral thrust.	Rod ko support chahiye. Slivers. Design aur manufacturing complex.
Slotted tube	Internal tube me lambai ki direction me slots (khanche) kate hote hain.	Neutral ke kareeb	Slots se stress aur thermal stress kam. Lambe grain (large L/D) ke liye acha. Neutral thrust ke kareeb.	Slivers bachte hain. Tooling aur processing complex.
Internal-burning star	Bore ki shape taare (star) jaisi, 5-8 points ke saath. Slide (Lecture 1, Page 13) me 5-slot grain isi family ka hai.	Neutral ke kareeb (contour lengths +-15% me)	Neutral thrust. Burning area bada aur high thrust. Stress kam star points ke curve se. Bahut widely used.	Star ke points par slivers. Stress concentration. Volumetric loading end-burning se kam.
External-burning rod	Solid rod jiski bahar ki surface jalti hai.	Regressive (A_b ghatta hai)	Simple. Shuru me high thrust milta hai.	Case ko garam gas se protection nahi milti. Thrust time ke saath girta hai. Practical motors me kam use hota hai.
Complex shapes: wagon wheel, dendrite, finocyl, conocyl	Multiple slots, fins ya cones wali complicated cross-section.	Design se tailored (programmed) thrust curve milta hai	High volumetric loading. Thrust-time curve ko design ke hisaab se shape kar sakte hain. Bade motors me stress aur slivers manage karne ke liye.	Design, analysis aur casting mehnga aur complex.

Yaad rakho

Exam me ye format rakho: **kaisi shape - kaisa burning (progressive/neutral/regressive) - 2 advantages - 2 limitations**. Aur ye line zaroor likho: grain geometry motor requirements (thrust-time curve, volume, C.G.) ke hisaab se choose hoti hai (Lecture 1, Page 24-26).

Grain design ki requirements (slide se)

Source: RP UNIT 3 - Lecture 1.pdf | Page 24 se 28

- **Motor requirements:** total impulse, thrust-time curve aur tolerance, motor mass, storage/operating temperature limits, available vehicle volume, C.G. location, vehicle forces se aane wali accelerations.
- **Geometry:** compact ho, burn surface vs time profile desired thrust-time curve se match kare, erosive burning avoid ya control ho, slivers aur C.G. shift kam ho.
- **Structural integrity:** grain, liner aur insulator loading, acceleration aur thermal stress me fail na ho.
- **Internal cavity:** perforations, slots, ports aur fins ki cavity burn time ke saath badhti hai, isliye resonance, vibration damping aur combustion stability check karni padti hai.
- **Processing:** grain ki processing repeatable, simple aur low cost ho.

EXAM ANSWER (write this in English for full marks)

Q. Explain the different solid propellant grain configurations with their geometrical characteristics, advantages and limitations.

A **grain** is the shaped block of solid propellant inside the motor. Its geometry decides how the burning surface area A_b changes with time, and therefore the thrust-time curve. The geometry is selected to match the mission requirements: total impulse, thrust-time curve, available volume and centre of gravity.

1. End-burning grain

- **Geometry:** solid cylinder; only one end face burns, axially. $A_b = \pi D^2/4$ = constant, so burning is neutral.
- **Advantages:** simple; constant thrust; long burn time; high volumetric loading; the unburnt propellant keeps the case cool.
- **Limitations:** low thrust because A_b is small; C.G. shifts during burning; the case end gets heated; bond stresses.

2. Internal-burning tube (cylindrical bore)

- **Geometry:** cylinder with a central circular hole; the bore surface burns radially outward. $A_b = \pi dL$ increases with time, so burning is progressive.
- **Advantages:** simple and common; high thrust possible; the propellant protects the case from hot gas.
- **Limitations:** thrust and pressure rise with time; slivers remain; erosive burning in the bore.

3. Rod and tube

- **Geometry:** a tube with a rod at its centre; the inner surface of the tube and the outer surface of the rod both burn. One surface grows while the other shrinks, so burning is nearly neutral.
- **Advantages:** large burning area in a small size, hence high thrust.
- **Limitations:** the rod needs a support; slivers; complex design and manufacture.

4. Slotted tube

- **Geometry:** internal tube with longitudinal slots; nearly neutral burning.
- **Advantages:** slots reduce stress and thermal stress; suitable for long grains.
- **Limitations:** slivers; complex tooling.

5. Internal-burning star

- **Geometry:** bore shaped like a star with 5 to 8 points; nearly neutral (burning contour lengths within +-15%).
- **Advantages:** neutral thrust, large burning area, low stress because of the rounded points; widely used.
- **Limitations:** slivers at star points; stress concentration; lower volumetric loading than end-burning.

6. External-burning rod

- **Geometry:** solid rod burning on its outer surface; A_b decreases, so burning is regressive.
- **Advantages:** simple; high thrust at the start.
- **Limitations:** case is not protected from hot gas; thrust falls with time; rarely used.

7. Complex shapes (wagon wheel, dendrite, finocyl, conocyl)

- Used to tailor the thrust-time curve and to get high volumetric loading in large motors. They are costly and complex to design and cast.

Conclusion: grain selection is a compromise between thrust level, burn time, volumetric loading, slivers, structural stress and cost. (Draw neat cross-section sketches of each grain, marking the burning surface, for full marks.)

EXAM ANSWER (write this in English for full marks)

Q. State the requirements to be satisfied in solid propellant grain design.

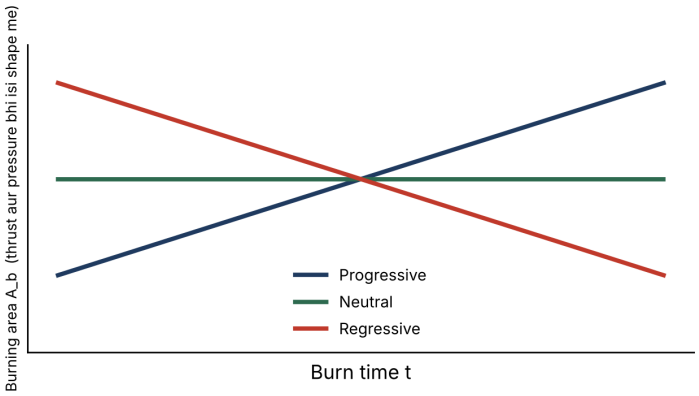
Grain design has to satisfy several interrelated requirements:

- **Motor requirements** from the flight mission: total impulse, desired thrust-time curve and tolerance, motor mass, storage and operating temperature limits, available vehicle volume, C.G. location and accelerations from vehicle forces.
- **Grain geometry:** compact, uses the available volume efficiently, burning surface versus time matches the required thrust-time curve, erosive burning is avoided or controlled, and slivers and C.G. shift are minimised.
- **Structural integrity:** the grain, liner and insulator must not fail from stress or strain under loading, acceleration or thermal stress. The geometry can be changed to reduce stress.
- **Internal cavity:** the volume of perforations, slots, ports and fins grows with burn time, so it must be checked for resonance, vibration damping and combustion stability.
- **Processing:** grain processing and propellant fabrication must be repeatable, simple and low cost, with acceptable thermal stress.

2. Progressive, neutral aur regressive burning

Source: Lecture 1 | Page 13-14 (burning surface contours, +-15% = neutral, A_b girne par thrust aur pressure girte hain). Naam aur examples: textbook (Sutton)

Ye teeno naam is baat se aate hain ki **burning surface area A_b time ke saath kaise badal raha hai**. Kyunki mass generation rate $\dot{m} = \rho A_b r$ aur chamber pressure A_b ke saath badhta hai, isliye A_b ki shape hi thrust ki shape hoti hai.



Type	A_b time ke saath	Thrust / pressure	Example grain
Progressive	Badhta hai	Time ke saath badhta hai	Internal-burning tube (bore bada hota hai)
Neutral	Lagbhag constant (+-15% ke andar)	Lagbhag constant	End-burning, star, rod and tube, slotted tube
Regressive	Ghatta hai	Time ke saath girta hai	External-burning rod

- **Neutral burning** sabse zyada chahiye hota hai kyunki thrust constant rehta hai. Slide ke hisaab se contour lines ki lengths +-15% ke andar hon to area constant maana jaata hai.
- End me A_b bahut girta hai to thrust aur pressure gir jaate hain. Pressure ek minimum se neeche aaye to combustion ruk jaata hai aur slivers bach jaate hain.
- Progressive grain me end tak pressure badhta jaata hai, isliye case ko zyada pressure ke liye design karna padta hai.
- Grain ko mission ki thrust-time requirement ke hisaab se choose karte hain. Kabhi shuru me high thrust chahiye (regressive), kabhi end me (progressive).

EXAM ANSWER (write this in English for full marks)

Q. Explain progressive, neutral and regressive burning of solid propellant grains.

The type of burning depends on how the burning surface area A_b changes with time. The mass generation rate is $\dot{m} = \rho_b A_b r$ and the chamber pressure is $p_c = K\rho_b rC^*$, with $K = A_b/A_t$. So thrust and chamber pressure follow the same trend as A_b.

- **Progressive burning:** A_b increases with time, so thrust and chamber pressure increase. Example: internal-burning tube (bore grows).
- **Neutral burning:** A_b stays almost constant (contour lengths within +-15%), so thrust and pressure stay almost constant. Examples: end-burning, star, rod and tube.
- **Regressive burning:** A_b decreases with time, so thrust and pressure fall. Example: external-burning rod.

Neutral burning is usually preferred for constant thrust. Near burn-out A_b falls, pressure drops, and below a minimum pressure combustion stops, leaving unburnt propellant (slivers). The type of burning is chosen from the thrust-time curve required by the mission. (Draw the A_b versus time graph with three curves.)

3. Burn rate ko influence karne wale factors

Source: RP UNIT 3 - Lecture 1.pdf | Page 15 (general 5 factors), Page 16-19 (composite ke 5 tareeke); RP UNIT 3 - Lecture 3.pdf | Page 47-49 aur RP UNIT 3 - Lecture 4.pdf | Page 5-10 (pressure); RP UNIT 3 - Lecture 4.pdf | Page 11-15 (temperature); RP UNIT 3 - Lecture 4.pdf | Page 16-17 (variable n, erosive burning)

A. Kisi bhi propellant ke liye (Lecture 1, Page 15)

- **Start se pehle propellant ka temperature** zyada ho to burn rate badhta hai.
- **Chamber pressure** zyada ho to burn rate badhta hai.
- **Combustion gas ka temperature** zyada ho to burn rate badhta hai.
- **Burning surface ke parallel gas velocity** zyada ho to burn rate badhta hai (erosive burning).
- **Rocket motor ki motion:** acceleration aur spin se grain par stress aata hai, jisse burn rate badalta hai.

B. Composite propellant me burn rate badhane ke 5 tareeke (Lecture 1, Page 16-19)

- **Burning rate catalyst (modifier)** add karo ya uski percentage badhao. Ye propellant ka sirf 0.1 se 3.0% hota hai aur combustion zone ki chemical reactions ko tez karta hai.
- **Oxidizer particle size chhota karo.** Same mass ke liye chhote particles ka surface area bahut zyada, isliye mixing aur reaction tez, burn rate zyada.
- **Oxidizer percentage badhao.** Ye fuel, oxidizer, binder aur formulation par depend karta hai.
- **Binder ya plasticizer ki heat of combustion badhao.** Binder ingredients ko jodta bhi hai aur fuel bhi deta hai. Plasticizer processing/mechanical properties theek karta hai aur energy bhi de sakta hai.
- **Wires ya metal staples embed karo.** Metal heat ko aage ke propellant tak conduct karke use preheat kar deta hai.

C. Pressure aur temperature ka asar

- **Pressure:** burn rate law **$r = a\,p^n$**. Zyadatar propellants me n = 0.2 se 0.6. n bada ho to pressure ke chhote change se burn rate bahut badalta hai. n 1 ke paas jaaye ya 1 se zyada ho to pressure kuch milliseconds me khatarnak tareeke se badh sakta hai.
- **Ambient (initial) temperature:** typical composite propellant motors me chamber pressure 20 se 35% aur operating time 20 se 30% badal sakta hai. Plateau propellants me ek pressure range me burn rate lagbhag constant rehta hai, isliye temperature ka asar kam.
- **Temperature sensitivity:** burn rate ki sensitivity $\text{sigma}_p = (1/r)(dr/dT_b)$ at fixed p aur pressure ki sensitivity $\text{pi}_K = (1/p_1)(dp_1/dT_b)$ at fixed K.

- **Erosive burning:** burning surface ke upar gas ka high-velocity flow burn rate badha deta hai. Ye tab zyada hota hai jab port area throat area ke relative bada nahi hota (port-to-throat area ratio 4 ya kam).
- **Variable n:** n asal me constant nahi hota, p_1 (aur isliye T_b) ka function hota hai.

Factor	Badhne par burn rate
Initial propellant temperature	Badhta hai
Chamber pressure	Badhta hai (r = a p^n)
Combustion gas temperature	Badhta hai
Gas velocity parallel to surface	Badhta hai (erosive burning)
Catalyst (0.1 se 3%)	Badhta hai
Oxidizer particle size	Chhota karne par badhta hai
Oxidizer percentage	Formulation par depend
Binder/plasticizer heat of combustion	Badhta hai
Embedded metal wire	Badhta hai

EXAM ANSWER (write this in English for full marks)

Q. List and explain the factors that influence the burning rate of solid propellants.

The burning rate r is the rate at which the burning surface recedes, perpendicular to itself, in mm/s. For a given propellant it increases with:

- a **higher initial propellant temperature**;
- a **higher chamber pressure**, according to $r = a\,p^n$ (n is usually 0.2 to 0.6);
- a **higher combustion gas temperature**;
- a **higher gas velocity parallel to the burning surface** (erosive burning, likely when the port-to-throat area ratio is 4 or less);
- **motor motion**: acceleration and spin-induced grain stress.

For composite propellants the burning rate can be increased by:

- adding a **burning rate catalyst** (0.1 to 3% of the propellant), which accelerates the reactions in the combustion zone;
- **decreasing the oxidizer particle size** (more surface area, faster mixing and reaction);
- **increasing the oxidizer percentage** (depends on the fuel, oxidizer, binder and formulation);
- **increasing the heat of combustion of the binder or plasticizer**;
- **embedding metal wires or staples**, which conduct heat ahead of the burning surface and preheat the propellant.

Effect of n: a high n makes burning rate and pressure very sensitive to each other; if n approaches 1, chamber pressure can rise dangerously in milliseconds, so $n < 1$ is needed for stable operation. Plateau propellants show nearly constant burning rate over a limited pressure range and reduce the effect of initial temperature.

4. Burning rate, burning surface area aur mass generation rate ka relation

Source: RP UNIT 3 - Lecture 3.pdf | Page 42 se 46 (mass flow relations); RP UNIT 3 - Lecture 3.pdf | Page 47 (r = a p^n); RP UNIT 1 - Lecture 1.pdf | Page 29 (equations r_b = a p_c^n aur m_dot = rho_p A_b r)

Burning surface se jo gas har second banti hai, wo teen cheezon ka product hai: burning area, burning rate aur propellant density:

$$\dot{m} = A_b\, r\, \rho_b$$

Jahan,

$\dot{m}$ = Propellant mass generation rate, kg/s

A_b = Burning surface area, m^2

r = Burning rate, m/s

ρ_b = Solid propellant density, kg/m^3

Ye gas chamber me store hoti hai aur nozzle se bahar nikalti hai (slide ka relation):

$$A_b\, r\, \rho_b \;=\; \frac{d(\rho_1 V_1)}{dt} \;+\; \frac{A_t\, p_1}{C^*}$$

Steady burning me storage wala term zero maan lete hain (preliminary design me ise ignore karte hain), to nozzle se nikalne wala flow hi generation rate ke barabar hota hai:

$$A_b\, r\, \rho_b \;=\; \frac{A_t\, p_1}{C^*} \quad\Rightarrow\quad p_1 \;=\; K\, \rho_b\, r\, C^*, \qquad K = \frac{A_b}{A_t}$$

Burn rate pressure par depend karta hai: r = a p_1^n. Dono ko jodne par steady chamber pressure milta hai:

$$p_1 \;=\; K\, \rho_b\, a\, p_1^{\,n}\, C^* \quad\Rightarrow\quad p_1 \;=\; (K\, \rho_b\, a\, C^*)^{\frac{1}{1-n}}$$

- A_b badhega to $\dot{m}$ badhega aur chamber pressure bhi badhega (K = A_b/A_t ke through).
- Burn rate r badhega (temperature ya pressure se) to $\dot{m}$ aur p_1 dono badhenge.
- Throat area A_t kam karoge to K badhega aur pressure badhega.
- Stable operation ke liye n < 1 hona zaroori hai. n >= 1 par pressure ka disturbance amplify hota hai (denominator 1 - n zero ya negative ho jaata hai).
- Preliminary calculation me A_t constant maante hain. Real firing me nozzle erosion se A_t 0.05 se 5% badh sakta hai.

Yaad rakho

Teen cheezein yaad rakho: **(1) m_dot = rho A_b r, (2) p = K rho r C*, (3) r = a p^n.** Inhi se burning area, burn rate, mass flow aur chamber pressure ke saare questions bante hain.

EXAM ANSWER (write this in English for full marks)

Q. Derive the relation between burning rate, burning surface area and propellant mass generation rate.

The gas produced per second by the burning surface is the product of burning area, burning rate and propellant density:

$\dot{m} = A_b\, r\, \rho_b$

where $\dot{m}$ is the mass generation rate (kg/s), A_b the burning area (m^2), r the burning rate (m/s) and ρ_b the propellant density (kg/m^3).

This mass is stored in the chamber and flows out through the nozzle: $A_b\, r\, \rho_b = d(\rho_1 V_1)/dt + A_t p_1/C^*$. For steady burning the storage term is neglected, so $A_b\, r\, \rho_b = A_t p_1/C^*$, which gives

$p_1 = K\, \rho_b\, r\, C^*$, where $K = A_b/A_t$

With the burning rate law r = a p_1^n, the steady chamber pressure is $p_1 = (K\, \rho_b\, a\, C^*)^{1/(1-n)}$.

- Increasing A_b or r increases the mass generation rate and the chamber pressure.
- Decreasing the throat area A_t increases K and the chamber pressure.
- Stable operation needs n < 1; for n ≥ 1 pressure disturbances are amplified.
- In preliminary design A_t is taken as constant (in real firing it increases by 0.05 to 5% due to nozzle erosion).

5. Problems me kaam aane wale formulas

Source: Slides ke solved numericals me use hue formulas: RP UNIT 3 - Lecture 1.pdf | Page 20-23, 30-31; RP UNIT 3 - Lecture 2 - Copy.pdf | Page 7-15; RP UNIT 3 - Lecture 3.pdf | Page 18-21

Kya nikalna hai	Formula	Symbols
Thrust (coefficient se)	$F = C_F\, p_c\, A_t$	C_F thrust coefficient, p_c chamber pressure, A_t throat area
Thrust (flow se)	$F = \dot{m}\, V_e$	V_e effective exhaust velocity
Specific impulse	$I_{sp} = F / \dot{m}$ (N.s/kg)	Numerically V_e ke barabar. Seconds me: I_{sp} / g_0, $g_0 = 9.81$
Exhaust velocity	$V_e = C_F\, C^* = I_{sp}$	C^* characteristic velocity
Characteristic velocity	$C^* = p_c\, A_t / \dot{m}$	
Mass flow	$\dot{m} = F / (C_F\, C^*) = F / I_{sp} = p_c\, A_t / C^* = \rho\, A_b\, r$	
Propellant mass	$m_p = \dot{m}\, t_b = I_t / I_{sp} = \rho\, A_b\, L$ (end-burning)	t_b burn time
Total impulse	$I_t = F\, t_b$	N.s
Burn time	$t_b = w / r$ (end-burning me L / r)	w web thickness
Chamber pressure	$p_c = \dot{m}\, C^* / A_t = K\, \rho\, r\, C^*$	$K = A_b / A_t$
Burn rate law	$r = a\, p^n$	a ke units n par depend karte hain
Rocket equation	$\Delta V = V_e \ln(m_0 / m_f)$	$m_0/m_f = e^{\Delta V/V_e}$

Problems - Solid Rocket

Slides me jitne bhi questions diye gaye the (total 6), un sab ko yahan collect karke solve kiya hai. Har question me order ye hai: **pehle formula, phir formula ke symbols ka matlab, phir given data, phir ek saath poori calculation** (numerator upar, denominator neeche).

Question 1

Source: RP UNIT 3 - Lecture 1.pdf | Page 20-21 (question + solution)

Ek propellant grain ki web thickness 40 mm hai aur wo 8 mm/s ke constant rate se jalta hai. Approximate burning time nikalo.

Formula: Burn time = web thickness / burning rate

$$t_b = \frac{w}{r}$$

Jahan,

t_b = Burning time, s

w = Web thickness, mm

r = Burning rate, mm/s

Ab, given data se hume pata hai ki

$$w = 40 \text{ mm} \qquad r = 8 \text{ mm/s}$$

Isliye,

$$t_b = \frac{40}{8}$$

$$t_b = 5 \text{ s}$$

$$\boxed{t_b \text{ (burning time)} = 5 \text{ s}}$$

Question 2

Source: RP UNIT 3 - Lecture 1.pdf | Page 22-23 (question + solution)

Ek single-stage rocket ko ΔV = 4500 m/s chahiye aur uska effective exhaust velocity V_e = 3000 m/s hai. Required initial-to-final mass ratio nikalo.

Formula: Rocket equation

$$\Delta V = V_e \ln\left(\frac{m_0}{m_f}\right) \qquad \Rightarrow \qquad \frac{m_0}{m_f} = e^{\Delta V/V_e}$$

Jahan,

ΔV	=	Rocket ko chahiye velocity change, m/s
V_e	=	Effective exhaust velocity, m/s
m_0	=	Initial mass (propellant ke saath), kg
m_f	=	Final mass (propellant jalne ke baad), kg

Given data se hume pata hai ki

$$\Delta V = 4500 \text{ m/s} \qquad V_e = 3000 \text{ m/s}$$

Isliye,

$$\frac{m_0}{m_f} = e^{4500/3000} = e^{1.5}$$

$$\frac{m_0}{m_f} = 4.48$$

$$\boxed{\frac{m_0}{m_f} \text{ (required)} = 4.48}$$

Question 3

Source: RP UNIT 3 - Lecture 1.pdf | Page 30-31 (question + given data) aur RP UNIT 3 - Lecture 2 - Copy.pdf | Page 5-11 (poora solution: ṁ p7, length p8-9, pressure p10, throat area p11)

Ek end-burning rocket cylindrical double-base propellant grain use karta hai jiska diameter 200 mm hai. Ye 300 s tak 350 N thrust deta hai. Thrust coefficient 1.15 hai. Propellant ki density = 1500 kg/m^3, a = 4 mm/s, n = 0.5, C* = 1500 m/s. Nikalo: (i) grain ki length, (ii) nozzle ka throat diameter.

Given data

- Grain diameter, D = 200 mm = 0.2 m
- Thrust, F = 350 N

- Burning time, t_b = 300 s
- Thrust coefficient, C_F = 1.15
- Propellant density, ρ_p = 1500 kg/m^3
- Burning rate coefficient, a = 4 mm/s = 0.004 m/s
- Pressure exponent, n = 0.5
- Characteristic velocity, C* = 1500 m/s

Step 1: Propellant mass flow rate

Formula:

$$\dot{m} = \frac{F}{C_F C^*}$$

Jahan,

$\dot{m}$	=	Propellant mass flow rate, kg/s
F	=	Thrust, N
C_F	=	Thrust coefficient
C^*	=	Characteristic velocity, m/s

Isliye,

$$\dot{m} = \frac{350}{1.15 \times 1500} = \frac{350}{1725}$$

$$\boxed{\dot{m} = 0.2029 \text{ kg/s}}$$

Step 2: Total propellant mass

Formula:

$$m_p = \dot{m}\, t_b$$

Jahan,

m_p	=	Total propellant consumed, kg
t_b	=	Burning time, s

$$m_p = 0.2029 \times 300$$

$$\boxed{m_p = 60.87 \text{ kg}}$$

Step 3: Burning area (end-burning grain)

End-burning grain me burning area, circular end ka area hota hai.

$$A_b = \frac{\pi D^2}{4}$$

Jahan,

A_b	=	Burning area, m^2

D = Grain diameter, m

$$A_b = \frac{\pi (0.2)^2}{4}$$

$$\boxed{A_b = 0.031416 \text{ m}^2}$$

Step 4: Burning rate aur grain length - (i) ka answer

Formula: mass flow ki relation se burning rate, aur phir length = burning rate x burning time

$$\dot{m} = \rho_p A_b r_b \Rightarrow r_b = \frac{\dot{m}}{\rho_p A_b} \qquad L = r_b \, t_b$$

Jahan,

r_b = Burning rate, m/s

ρ_p = Propellant density, kg/m^3

L = Grain ki length, m

$$r_b = \frac{0.2029}{1500 \times 0.031416} = 0.004306 \text{ m/s} = 4.306 \text{ mm/s}$$

$$L = r_b \, t_b = 0.004306 \times 300$$

$$\boxed{L \approx 1.29 \text{ m}}$$

Check: L = m_p / (ρ_p A_b) = 60.87 / (1500 x 0.031416) = 1.292 m, same aaya.

Step 5: Chamber pressure

Formula: burning rate law

$$r_b = a \, p_c^{\,n} \Rightarrow p_c = \left(\frac{r_b}{a}\right)^{1/n}$$

Jahan,

p_c = Chamber pressure

a = Burning rate coefficient

n = Pressure exponent

Dhyan do: a = 4 mm/s tabhi lagta hai jab pressure MPa me liya jaye (slide me bhi yahi liya gaya hai). Isliye p_c MPa me aayega.

$$p_c = \left(\frac{4.306}{4}\right)^{1/0.5} = (1.0765)^2$$

$$\boxed{p_c = 1.159 \text{ MPa}}$$

Step 6: Throat diameter - (ii) ka answer

Formula: thrust equation se throat area, phir area se diameter

$$F = C_F\, p_c\, A_t \;\Rightarrow\; A_t = \frac{F}{C_F\, p_c} \qquad\qquad A_t = \frac{\pi d_t^{\,2}}{4} \;\Rightarrow\; d_t = \sqrt{\frac{4A_t}{\pi}}$$

Jahan,

A_t	=	Throat area, m^2
d_t	=	Throat diameter, m
C_F	=	Thrust coefficient

$$A_t \;=\; \frac{350}{1.15 \times 1.159 \times 10^{6}} \;=\; 2.626 \times 10^{-4}\ \text{m}^2$$

$$d_t \;=\; \sqrt{\frac{4 \times 2.626 \times 10^{-4}}{\pi}} \;=\; 0.0183\ \text{m}$$

$$\boxed{d_t \;\approx\; 18.3\ \text{mm}}$$

Final answer

- (i) Grain ki length L = 1.29 m
- (ii) Throat diameter d_t = 18.3 mm

Question 4

Source: RP UNIT 3 - Lecture 2 - Copy.pdf | Page 12 (question), Page 13-15 (solution: pressure p13, burning rate p14, grain diameter p15)

Ek end-burning solid propellant rocket, jiska throat diameter 28 mm hai, missile ke sustainer ke roop me use hota hai. Use 5 kN ka constant thrust dena hai. Double-base propellant ka burn rate law r = 0.004 $P^{0.45}$ hai (P Pa me, r mm/s me). Specific impulse 1900 N.s/kg maan lo. Propellant ka C* = 1600 m/s aur density = 1540 kg/m^3 hai. Propellant grain ka diameter nikalo.

Given data

- Throat diameter, d_t = 28 mm = 0.028 m
- Thrust, F = 5 kN = 5000 N
- Burn rate law, r = 0.004 $P^{0.45}$ (P in Pa, r in mm/s)
- Specific impulse, I_{sp} = 1900 N.s/kg
- Characteristic velocity, C* = 1600 m/s
- Propellant density, ρ_p = 1540 kg/m^3

Step 1: Mass flow rate

Formula:

$$\dot{m} \;=\; \frac{F}{I_{sp}}$$

Jahan,

$\dot{m}$ = Propellant mass flow rate, kg/s

F = Thrust, N

I_{sp} = Specific impulse, N.s/kg

$$\dot{m} = \frac{5000}{1900}$$

$$\boxed{\dot{m} = 2.632 \text{ kg/s}}$$

Step 2: Throat area aur chamber pressure

Formula: C* ki definition se

$$C^* = \frac{p_c A_t}{\dot{m}} \Rightarrow p_c = \frac{\dot{m} C^*}{A_t} \qquad A_t = \frac{\pi d_t^2}{4}$$

Jahan,

p_c = Chamber pressure, Pa

A_t = Throat area, m^2

d_t = Throat diameter, m

$$A_t = \frac{\pi (0.028)^2}{4} = 6.158 \times 10^{-4} \text{ m}^2$$

$$p_c = \frac{2.632 \times 1600}{6.158 \times 10^{-4}}$$

$$\boxed{p_c = 6.839 \times 10^6 \text{ Pa}}$$

Step 3: Burning rate

Formula: given burn rate law

$$r = 0.004 P^{0.45}$$

Jahan,

r = Burning rate, mm/s

P = Chamber pressure, Pa

$$r = 0.004 (6.839 \times 10^6)^{0.45} = 0.004 \times 1190.6$$

$$\boxed{r = 4.762 \text{ mm/s} = 0.004762 \text{ m/s}}$$

Step 4: Burning area aur grain diameter

Formula: mass flow relation se burning area, phir area se diameter

$$\dot{m} = \rho_p A_b r \;\Rightarrow\; A_b = \frac{\dot{m}}{\rho_p\, r} \qquad\qquad A_b = \frac{\pi D_p^{\,2}}{4} \;\Rightarrow\; D_p = \sqrt{\frac{4A_b}{\pi}}$$

Jahan,

A_b	=	Burning area, m^2
ρ_p	=	Propellant density, kg/m^3
r	=	Burning rate, m/s
D_p	=	Propellant grain diameter, m

$$A_b = \frac{2.632}{1540 \times 0.004762} = 0.3589 \text{ m}^2$$

$$D_p = \sqrt{\frac{4 \times 0.3589}{\pi}} = 0.676 \text{ m}$$

$$D_p \approx 676 \text{ mm}$$

Slide ke answer me difference

Lecture slide me is question ke aage ke steps me burning rate **r = 8.90 mm/s** likha hai aur final diameter **495 mm** aaya hai. Lekin 0.004 x $(6.835 \times 10^6)^{0.45}$ calculator me daalne par r = **4.76 mm/s** aata hai, 8.90 nahi. Isi wajah se slide ka 495 mm galat lag raha hai; sahi value lagbhag **676 mm** hai. Exam me apna method aur calculator value hi likhna, aur sir se ek baar confirm kar lena.

Question 5

Source: RP UNIT 3 - Lecture 3.pdf | Page 18 (question), Page 19 (solution)

Ek solid rocket motor 6.5 MPa ke steady-state chamber pressure par chalta hai. Composite propellant ka empirical burn rate constant a = 4.8 mm/s.$(MPa)^{-n}$ aur pressure exponent n = 0.38 hai. (1) Is operating pressure par propellant ka linear burning rate r nikalo. (2) Agar propellant grain ek end-burning cylinder hai jiski initial length 450 mm hai, to motor ka total burn time (t_b) nikalo.

Given data

- Chamber pressure, p_c = 6.5 MPa
- Burn rate constant, a = 4.8 mm/s.MPa^{-n}
- Pressure exponent, n = 0.38
- Grain length (end-burning), L = 450 mm

Part 1: Burning rate

Formula: burn rate law

$$r = a\, p_c^{\,n}$$

Jahan,

r	=	Linear burning rate, mm/s
a	=	Burn rate constant, mm/s.MPa^{-n}

p_c = Chamber pressure, MPa

n = Pressure exponent

$$r = 4.8 \times (6.5)^{0.38} = 4.8 \times 2.0366$$

$$\boxed{r \approx 9.78 \text{ mm/s}}$$

Part 2: Burn time

End-burning grain me poori length L utni hi distance hai jo burning surface ko tay karni hai.

$$t_b = \frac{L}{r}$$

Jahan,

t_b = Burn time, s

L = Grain length, mm

r = Burning rate, mm/s

$$t_b = \frac{450}{9.776}$$

$$\boxed{t_b \approx 46.0 \text{ s}}$$

Note: slide me $(6.5)^{0.38}$ ko 2.0327 liya gaya hai, isliye wahan r = 9.76 mm/s aur t_b = 46.12 s aaya hai. Calculator se $(6.5)^{0.38}$ = 2.0366 aata hai, isliye maine 9.78 mm/s aur 46.0 s likha hai. Dono answers lagbhag same hain.

Final answer

- (1) r = 9.78 mm/s (slide: 9.76 mm/s)
- (2) t_b = 46.0 s (slide: 46.12 s)

Question 6

Source: RP UNIT 3 - Lecture 3.pdf | Page 20 (question), Page 21 (solution)

Ek chhota solid rocket motor 2.5 kg/s ka steady gas mass flow rate generate karta hai. Design chamber pressure P_c = 4.0 MPa (4.0 x 10^6 Pa) hai aur propellant exhaust gas ka characteristic velocity c* = 1,300 m/s hai. Is operating condition ko maintain karne ke liye required nozzle throat area (A_t) m^2 me nikalo.

Given data

- Mass flow rate, ṁ = 2.5 kg/s
- Characteristic velocity, c* = 1,300 m/s
- Chamber pressure, P_c = 4.0 MPa = 4.0 x 10^6 Pa

Formula: characteristic velocity ki definition se

$$c^* = \frac{p_c A_t}{\dot{m}} \qquad \Rightarrow \qquad A_t = \frac{\dot{m}\, c^*}{p_c}$$

Jahan,

A_t = Nozzle throat area, m^2

$\dot{m}$ = Mass flow rate, kg/s

c^* = Characteristic velocity, m/s

p_c = Chamber pressure, Pa

$$A_t = \frac{2.5 \times 1300}{4.0 \times 10^6} = \frac{3250}{4{,}000{,}000}$$

$$\boxed{A_t = 8.125 \times 10^{-4}\ \text{m}^2\ (= 812.5\ \text{mm}^2)}$$